higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

APPLIED ENGINEERING TECHNOLOGY
NQF LEVEL 4

(6021024)

27 February 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL the sketches must be neat and in proportion.
5. $g = 9{,}81\ m/s^2$
6. Write neatly and legibly.

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QUESTION 1

1.1 1.1.1 Differentiate between *kinetic energy* and *potential energy.* (4)

1.1.2 Define *centrifugal force.* (2)

1.1.3 Explain how the use of different lengths of levers, when applied to a force acting around a point, will influence torque acting on that point. (4)

1.2 List FIVE types of gears used in modern engineering systems. (5)

1.3 What is a hydraulic accumulator? (2)

1.4 John is pulling a cardboard box full of books on a horizontal surface. The box has a mass of 80 kg and it requires a force of 180 N to pull it at constant speed.

Calculate the following:

1.4.1 The coefficient of friction (4)

1.4.2 The angle of friction (3)

1.4.3 The horizontal force that will be required if a similar box is placed on top of the first one. (4)

[28]

QUESTION 2

A beam with a length of 8 meters' rests on two supports which are 5 meters apart from each other. The support on the right side is 2 meters away from the nearest end. This beam is bearing a uniformed distribution load (UDL) of 408kg/m between the two supports, a concentrated load of 1 kN at the left end and a concentrated load of 2 kN at the right end of the beam.

2.1 Use a scale of 1 cm = 1 m and draw the beam. Indicate ALL forces on the diagram. (3)

2.2 Calculate the reactions at the supports and test your answer. (7)

2.3 Use a scale of 1 mm = 100 N and draw the shear-force diagram. Indicate ALL values on the diagram. (5)

2.4 Calculate the bending moments at points A, B, C, D and E. (5)

[20]

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QUESTION 3

3.1 *Control systems engineering* is an advanced field of engineering that is concerned with analysing systems, identifying the problem that they may be experiencing, and the design and construction of a control system to solve the problem.

3.1.1 What is computer-aided design software? (2)

3.1.2 Name TWO types of control systems. (2)

3.1.3 Explain FOUR impacts of a poor design in construction and manufacturing industries. (4)

3.2 Calculate the height that the water will rise in a pipe system of a building if the gauge pressure on the ground floor registers 350 kPa. (6)

3.3 Machines used in industry are potentially dangerous if not operate properly.

Explain FIVE basic situations where machines are most dangerous. (5)

3.4 Explain the function of each of the following components:

3.4.1 Oil pump

3.4.2 Oil filter

3.4.3 Oil sump

3.4.4 Dipstick

(4 × 1) (4)

3.5 State TWO hazards that may occur in a workshop. (2)
[25]

QUESTION 4

4.1 Calculate the pressure on a surface area of 3 000 mm^2 if the pressure force is 0,007 kN. Give the answer in kPa.

(HINT: $Pressure = \frac{F}{A}$) (5)

4.2 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–K) next to the question number (4.2.1–4.2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.2.1	Control valve	A	a device which conducts current between stationary wires and moving parts, most commonly in a rotating shaft
4.2.2	Engine cylinder	B	a nearly incompressible fluid that conforms to the shape of its container but retains a (nearly) constant volume independent of pressure
4.2.3	OHSA	C	fitted to the motor where the driving motion of the V-belt starts
4.2.4	Liquid	D	a valve used to control fluid flow by varying the size of the flow passage as directed by a signal from a controller
4.2.5	Viscosity	E	brings the moving angular velocity to rest
4.2.6	Drive pulley	F	a first or preliminary version of a device or vehicle from which other forms are developed
4.2.7	Brush	G	occupational health and safety act
4.2.8	Electric motor	H	the central working part of a reciprocating engine or pump, the space in which a piston travels
4.2.9	Brake	I	converts electrical energy to mechanical energy
4.2.10	Prototype	J	the speed of flow or resistance to flow or the 'thicknesses of a fluid
		K	converts mechanical energy to electrical energy

(10 × 1) (10)

[15]

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QUESTION 5

5.1 List FOUR types of motions. (4)

5.2 A servo motor can be either DC or AC.

Describe the operation of a servo motor (4)

5.3 Name THREE types of impellers that can be used on centrifugal pumps. (3)

5.4 State the function of an evaporator. (1)

[12]

TOTAL: 100

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